

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0307

East Bismarck 100 kV
February 28, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0307. The Interconnection Customer proposes a 47.1 MW generating facility interconnecting at the East Bismarck 100 kV substation in the Illinois area. The request was received on February 08, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 213 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 10 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,362,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of TEXON to E BISMARCK 230 KV L, the E BISMARCK to RIO BRAVO ROCKLIN 100 KV Circuit 1 loads to 108.7% of its 151 MVA rating. Project DFAX: 26.7%.

For the loss of TEXON to E BISMARCK 230 KV line, the E BISMARCK to RIO BRAVO ROCKLIN 100 KV #2 loads to 108.7% of its 151 MVA rating. Project DFAX: 26.7%.

For the loss of PINEY CREEK PROJECT to HILLTOP 500 KV, the PINEY CRK PROJECT 500/345kV Transformer loads to 108.0% of its 1,548 MVA rating. Project DFAX: 9.2%.

FISHKILL PLAINS to DILLARD-ORANGE 345 KV line: 106.6% loading (1,913 MVA) following the outage of the TIDD to MUSKINGUM RIV 500 KV L; DFAX = 5.8%.

The CARBON CTR 500 kV/345 kV Auto is overloaded to 106.0% of its 1,187 MVA emergency rating for the loss of the BLACKROCK to SHWVILLE STA 345 KV. The project contributes 6.9% DFAX (3.2 MW) to this overload.

LIONS ROAD to PINEY CRK PROJECT 500 KV: 103.1% loading (2,671 MVA) following the outage of the EL PASO SUBSTATION to GANSO 500 KV; DFAX = 13.3%.

CALDWELL to PINEY CRK PROJECT 345 KV: 102.9% loading (1,621 MVA) following the outage of the HILLTOP to PINEY CRK PROJECT 500 KV; DFAX = 9.3%.

HILLTOP to STRAWBERRY BELT 345 KV L: 102.4% loading (1,371 MVA) following the outage of the HILLTOP to TALENENERGY MARTINS CRK LLC WILLIAMSPORT 500 KV; DFAX = 6.1%.

BURWELL 500/345 kV XFMR: 102.0% loading (856 MVA) following the outage of the ALLEGHENY ENERGY UNITS 3 4 & 5 345/1 kV Auto; DFAX = 6.4%.

CARBON CTR to CALDWELL 345 KV Line: 101.8% loading (1,074 MVA) following the outage of the BLACKROCK to SHAWVILLE STA 345 KV; DFAX = 5.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
E BISMARCK to RIO BRAVO ROCKLIN 100 KV Circuit 1	Reconductor line	\$2,010,000	19% (\$379,000)
EAST BISMARCK to RIO BRAVO ROCKLIN 100 KV	Reconductor line	\$1,151,000	19% (\$217,000)
PINEY CRK PROJECT 500-345 kV XFMR	Replace transformer with higher-rated unit	\$38,186,000	1% (\$250,000)
FISHKILL PLAINS to DILLARD-ORANGE 345 KV L	Reconductor line	\$56,486,000	1% (\$365,000)
CARBON CTR 500 KV/345 KV Auto	Replace transformer with higher-rated unit	\$34,802,000	1% (\$228,000)
LIONS ROAD to PINEY CRK PROJECT 500 KV	Reconductor line	\$66,269,000	1% (\$434,000)
CALDWELL to PINEY CRK PROJECT 345 KV	Reconductor line	\$27,380,000	1% (\$179,000)
HILLTOP to STRAWBERRY BELT 345 KV L	Reconductor line	\$13,941,000	1% (\$91,000)
BURWELL 500/345 kV XFMR	Replace transformer with higher-rated unit	\$17,568,000	1% (\$114,000)
CALDWELL to CARBON CENTER 345 KV	Reconductor line	\$15,985,000	1% (\$105,000)
	Total allocated to project		\$2,362,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.